

A Complexity Analysis.

Given a directed graph $G = (V, E)$, we evaluate the worst-case running time of Algorithm 4. Let $T(m, n)$ be the worst-case running time of function BK2(G, k_1, k_2, q, P, C, X) where $m = |C|$ and $n = |P \cup C|$, and we define $k = \max(k_1, k_2)$. Next, we analyze $T(m, n)$ by considering four different cases:

I. When $C = \emptyset$ (Lines 4–6), then

$$T(m, n) = T(0, n) \leq c_1, \quad (4)$$

where c_1 is a constant.

II. When $P \cup C$ is not a (k_1, k_2) -plex and the pivot vertex is in P (Lines 8–17). Here, $p + 1$ branches are created each with a smaller candidate set C' . Before each call the sets C' and X' are further shrunk in Lines 15–16 which takes time $O(|V|^2)$. The worst-case running time of this case is, therefore

$$T(m, n) \leq (p + 1)c_2|V|^2 + \sum_{i=1}^p T(m - i, n - 1) + T(m - d, n - d + p), \quad (5)$$

where c_2 is a constant.

III. When $P \cup C$ is a (k_1, k_2) -plex (Lines 19–24), then no further branches are created, and the cost in this case is only incurred by updating X' in Line 22, which is

$$T(m, n) \leq c_2|V|^2 \quad (6)$$

IV. When $P \cup C$ is not a (k_1, k_2) -plex and the pivot vertex is in C (Lines 25–30). The cost in this case is

$$T(m, n) \leq c_2|V|^2 + T(m - 1, n) + T(m - 1, n - 1).$$

In the worst case, no candidate vertex is reduced from $C - \{v_p\}$ in Line 28 so v_p remains the minimum degree vertex in the subsequent recursive call at Line 30 (within which $v_p \in P' = P \cup \{v_p\}$). Hence, by expanding $T(m - 1, n)$ using Equation (5) we have:

$$T(m, n) \leq (p + 2)c_2|V|^2 + \sum_{i=1}^p T(m - 1 - i, n - 1) + T(m - 1 - d, n - d + p) + T(m - 1, n - 1). \quad (7)$$

Next, we compute a closed-form upper-bound of $T(m, n)$ which satisfies the Equations (4)–(7), and we can drop the argument n from $T(m, n)$ since the base case expressions in Equations (4) and (6) are not a function of n . Then, the largest $T(m)$ is upper-bounded by Equations (4) and (7), i.e.

$$T(m) \leq \begin{cases} c_1 & \text{if } m = 0 \\ (p + 2)c_2|V|^2 + \sum_{i=0}^p T(m - 1 - i) & \\ +T(m - 1 - d) & \text{otherwise} \end{cases} \quad (8)$$

Since $p \leq d - 1$ and $p < k$, the worst-case is reached when $p = k - 1$ and $d = k$. Then, Equation (8) can be rewritten as:

$$T(m) \leq \begin{cases} c_1 & \text{if } m = 0 \\ (k + 1)c_2|V|^2 + \sum_{i=0}^k T(m - 1 - i) & \text{otherwise} \end{cases} \quad (9)$$

We show that $T(m) = O(\alpha_k \gamma_k^m - \beta_k)$ where γ_k is largest real root of $x^{k+2} - 2x^{k+1} + 1 = 0$, $\beta_k = \frac{(k+1)c_2|V|^2}{k}$ and $\alpha_k = \frac{(2^k k + 1)\beta_k + 2^k c_1}{\gamma_k^{k+1}}$.

LEMMA A.1. Let $H(m) = \alpha_k \gamma_k^m - \beta_k$ where γ_k is largest real root of $x^{k+2} - 2x^{k+1} + 1 = 0$, $\beta_k = \frac{(k+1)c_2|V|^2}{k}$ and $\alpha_k = \frac{(2^k k + 1)\beta_k + 2^k c_1}{\gamma_k^{k+1}}$. If $T(m)$ satisfies Equation (9), then $T(m) \leq H(m)$ for $m \geq 0$.

PROOF. We first show $T(m) \leq 2^{m-1}k\beta_k + 2^{m-1}c_1$ by induction on m .

Basis: for $m = 1$, $T(1) \leq (k + 1)c_2|V|^2 + T(0) \leq k\beta_k + c_1$.

Induction: We assume that $T(i) \leq 2^{i-1}k\beta_k + 2^{i-1}c_1$ for all $i = 1, \dots, m - 1$ and prove that it also holds for $i = m$. By Equation (9) we have:

$$\begin{aligned} T(m) &\leq (k + 1)c_2|V|^2 + T(m - 1) + \dots + T(0) \\ &= (k + 1)c_2|V|^2 + c_1 + \sum_{i=1}^{m-1} T(i) \\ &\leq (k + 1)c_2|V|^2 + c_1 + \sum_{i=1}^{m-1} (2^{i-1}k\beta_k + 2^{i-1}c_1) \quad (10) \\ &= k\beta_k(1 + \sum_{i=1}^{m-1} 2^{i-1}) + c_1(\sum_{i=1}^{m-1} 2^{i-1} + 1) \\ &= 2^{m-1}k\beta_k + 2^{m-1}c_1 \end{aligned}$$

Then, we prove that $T(m) \leq H(m)$ for all $m \geq 0$ again by induction on m .

Basis: for $m = 1$, $T(1) \leq (k + 1)c_2|V|^2 + T(0) \leq k\beta_k + c_1$, and $H(1) = \alpha_k \gamma_k - \beta_k = \frac{(2^k k + 1)\beta_k + 2^k c_1}{\gamma_k^k} \geq k\beta_k + c_1$ (since $\gamma_k < 2$ and hence $\gamma_k^k < 2^k$).

Induction: We assume that $T(m) \leq H(m)$ for $m = 1, \dots, i$ and prove that it also holds for $m = i + 1$. First, let us factorize $x^{k+1} - 2x^{k+1} + 1$ as $(x - 1)(x^{k+1} - x^k - \dots - 1)$. Denote $F_k(x) = (x^{k+1} - x^k - \dots - 1)$. We have $F_k(1) < 0$ and $F_k(2) > 0$, indicating that $F_k(x) = 0$ has a root between $(1, 2)$. Combining the fact that γ_k is also the root of $F_k(x) = 0$, thus $\gamma_k^k + \dots + 1 = \gamma_k^{k+1}$. Multiply both side with γ_k^{i-k} , we have $\gamma_k^i + \dots + \gamma_k^{i-k} = \gamma_k^{i+1}$. Then the following inequalities hold.

$$\begin{aligned} T(i + 1) &\leq T(i) + \dots + T(i - k) + (k + 1)c_2|V|^2 \\ &\leq \alpha_k(\gamma_k^i + \dots + \gamma_k^{i-k}) - (k + 1)\beta_k + k\beta_k \\ &\leq \alpha_k(\gamma_k^i + \dots + \gamma_k^{i-k}) - \beta_k \quad (11) \\ &= \alpha_k \gamma_k^{i+1} - \beta_k \\ &= H(i + 1) \quad \square \end{aligned}$$

Finally, as we call BK2(.) with $m = |V|$ and k is a constant, we get the worst-case running time of our algorithm as given in Theorem 6.1.

B Complete Results in Parallel Executions

We implemented the parallel version of Ours based on the description in Section 8. Table 5 reports the (k_1, k_2) -plex enumeration time of parallel Ours on all our datasets in Table 1 when running with 2, 4, 8, 16 and 32 threads, respectively, where we use the default

Table 5: The Running Time of Mining Large Maximal (k_1, k_2) -Plexes in Parallel Execution

Dataset	k_1	k_2	q	$\#(k_1, k_2)$ - plexes	Serial	2 Threads	4 Threads	8 Threads	16 Threads	32 Threads
bitcoin	2	5	10	144.00	0.06	0.04	0.02	0.01	0.01	0.02
			12	0	0.01	0.00	0.00	0.00	0.00	0.00
	3	5	10	24,261	8.35	4.78	2.46	1.29	0.66	0.36
			12	286	1.05	0.62	0.33	0.17	0.09	0.05
wiki-vote	3	5	10	474	1.85	1.08	0.55	0.29	0.15	0.08
			12	0	0.08	0.05	0.02	0.01	0.01	0.01
	4	5	10	30,222	349.32	197.94	102.45	52.57	26.57	16.44
			12	94	16.63	9.76	4.99	2.61	1.32	0.91
mathoverflow	2	2	10	523,633	28.55	18.48	9.48	4.93	2.50	1.34
			12	150,888	13.35	9.01	4.56	2.38	1.22	0.69
	2	3	10	1,762,917	63.16	47.81	24.41	12.71	6.45	3.52
			12	602,862	32.44	24.92	12.61	6.63	3.36	1.86
as-caida	2	3	10	23,314	7.99	0.34	0.19	0.10	0.08	0.15
			15	185	0.93	0.02	0.01	0.01	0.01	0.02
	3	4	15	17,303	14.87	0.63	0.33	0.18	0.10	0.10
			20	0	0.19	0.01	0.00	0.00	0.00	0.00
epinions	2	3	10	773,095	23.85	18.34	9.48	4.94	2.56	1.41
			12	252,511	10.33	8.06	4.18	2.23	1.19	0.80
	3	4	12	24,599,029	2,053.04	1,174.05	605.69	314.25	158.78	85.62
			15	2,382,084	381.99	229.27	118.32	61.53	31.62	18.13
email-euall	2	3	10	17,438	1.69	1.26	0.65	0.34	0.18	0.19
			12	1442	0.58	0.39	0.21	0.11	0.07	0.12
	3	4	12	196,878	80.58	46.44	23.49	12.18	6.14	3.38
			15	1351	9.19	5.20	2.68	1.40	0.72	0.44
amazon0505	3	5	10	4,696	0.27	0.08	0.04	0.02	0.02	0.01
			12	24	0.14	0.01	0.00	0.00	0.00	0.00
	5	3	10	1,532	0.20	0.04	0.02	0.01	0.01	0.00
			12	1	0.13	0.00	0.00	0.00	0.00	0.00
web-google	2	3	15	997	0.29	0.05	0.02	0.01	0.01	0.00
			20	57	0.21	0.01	0.01	0.00	0.00	0.00
	3	4	15	1,993	0.32	0.06	0.03	0.02	0.01	0.01
			20	61	0.22	0.02	0.01	0.00	0.00	0.00
soc-pokec	2	3	15	17,980	2.11	0.92	0.49	0.25	0.21	0.39
			20	17	0.95	0.03	0.01	0.01	0.01	0.02
	3	3	15	508,884	54.91	32.24	16.52	8.65	4.41	2.41
			20	648	1.66	0.49	0.25	0.13	0.07	0.06
wiki-talk	2	2	15	22,219	287.28	173.50	87.73	44.75	22.50	11.48
			20	0	69.70	38.84	19.89	10.27	5.17	2.67
	2	3	15	165,156	525.13	376.12	191.71	97.82	48.99	25.01
			20	0	129.02	77.30	39.23	20.22	10.16	5.21
arabic-2005	2	2	1000	106,895	3,906.70	1,897.35	1,017.31	527.40	269.04	140.58
	3	3	3000	3,500	111.60	60.20	29.56	15.10	8.69	4.64
uk-2005	2	2	200	117,255	301.09	162.70	84.11	43.34	21.82	11.38
	3	3	400	66,639	168.98	91.51	46.34	23.95	12.11	6.34
it-2004	2	2	1000	15,087	1,170.05	621.80	321.12	166.53	85.40	52.04
	3	3	3000	1,034	77.90	41.35	21.72	10.21	6.40	3.94
webbase-2001	2	2	300	327,882	646.41	320.84	164.46	90.02	53.56	30.75
	3	3	500	88,777	184.44	95.16	52.07	30	18.15	8.75
clue-web	2	2	10	1,215,977	26.05	14.15	7.66	3.82	1.75	1.69
	3	3	20	95,383	52.92	31.76	16.46	8.56	4.42	2.37

timeout threshold $\tau_{time} = 0.1$ ms is adopted, which is tested to work consistently well on various datasets.

As Table 5 shows, our parallel algorithm is efficient and scales up well with the number of CPU cores on all the datasets.